

Course Code	21CSC202J	Course Name	OPERATING SYSTEMS		Course Category	C	PROFESSIONAL CORE				L	T	P	C
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Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards	Nil		Nil

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	outline the structure of OS and basic architectural components involved in OS design
CLR-2:	introduce the concept of deadlock and various memory management mechanism
CLR-3:	familiarize the scheduling algorithms, file systems, and I/O schemes
CLR-4:	identify and tell the various embedded operating systems and computer security concepts
CLR-5:	name the various computer security techniques in windows and Linux

Course Outcomes (CO):	At the end of this course, learners will be able to:
CO-1:	use the appropriate concepts of operating system for resource utilization
CO-2:	choose the relevant process and thread concepts for solving synchronization problems
CO-3:	exemplify different types of scheduling algorithms and deadlock mechanism
CO-4:	experiment the performance of different algorithms used in management of memory, file and I/O and select the appropriate one
CO-5:	demonstrate different device and resource management techniques for memory utilization with security mechanisms

Unit-1 - Introduction	15 Hour
Computer-System Organization, Computer-System Architecture, Operating-System Structure, Operating-System Operations, Process Management, Memory Management, Storage Management, Protection and Security, Kernel Data Structures, Computing Environments, Open-Source Operating Systems, Operating-System Services, User and Operating-System Interface, System Calls, Types of System Calls, System Programs, Operating-System Design and Implementation, Operating-System Structure, Operating-System Debugging, Operating-System Generation, System Boot.	
Unit-2 - Process Management	15 Hour
Process Concept, Process Scheduling, Operations on Processes, Interprocess Communication, Communication in Client- Server Systems, Threads: Multicore Programming, Multithreading Models, Thread Libraries, Implicit Threading, Threading Issues. Process Synchronization: The Critical-Section Problem, Peterson's Solution, Synchronization Hardware, Mutex Locks, Semaphores, Classic Problems of Synchronization, Monitors	
Unit-3 - CPU Scheduling	15 Hour
Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Thread Scheduling, Multiple-Processor Scheduling, Real-Time CPU Scheduling. Deadlocks: System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock	
Unit-4 - Memory Management	15 Hour
Main Memory, Swapping, Contiguous Memory Allocation, Segmentation, Paging, Structure of the Page Table. Virtual Memory: Introduction, Demand Paging, Copy-on-Write, Page Replacement, Allocation of Frames, Thrashing, Memory-Mapped Files, Allocating Kernel Memory. STORAGE MANAGEMENT: Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, Swap-Space Management, RAID Structure. File-System Interface: File Concept, Access Methods, Directory and Disk Structure, File-System Mounting, File Sharing, Protection. .	

15 Hour

Goals of Protection, Principles of Protection, Domain of Protection, Access Matrix, Implementation of the Access Matrix, Access Control, Revocation of Access Rights, Capability-Based Systems, Language-Based Protection, The Security Problem, Program Threats, System and Network Threats, Cryptography as a Security Tool, User Authentication, Implementing Security Defenses, Firewalling to Protect Systems and Networks, Computer-Security Classifications.

Lab Experiments

Lab 1: Operating system Installation, Basic Linux commands	Lab 6: Dining Philosopher problem	Lab 11: LRU and LFU Page Replacement Algorithm
Lab 2: Process Creation using fork() and Usage of getpid(), getppid(), wait() functions	Lab 7: Bankers Algorithm for Deadlock avoidance	Lab 12: Best fit and Worst fit memory management policies
Lab 3: Multithreading	Lab 8: FCFS and SJF Scheduling	Lab 13: Disk Scheduling algorithm
Lab 4: Mutual Exclusion using semaphore and monitor	Lab 9: Priority and Round robin scheduling	Lab 14: Sequential and Indexed file Allocation
Lab 5: Reader-Writer problem	Lab 10: FIFO Page Replacement Algorithm	Lab 15: File organization schemes for single level and two-level directory

Learning Resources	1. Abraham Silberschatz, Peter Baer Galvin, and Greg Gagne, "Operating System Concepts", John Wiley & Sons (Asia) Pvt. Ltd, Tenth Edition, 2018 2. RamazElmasri, A. Gil Carrick, David Levine, "Operating Systems – A Spiral Approach ", Tata McGraw Hill Edition, 2010 3. Dhnananjay M. Dhamdhare, "Operating Systems – A Concept Based Approach", Third Edition, Tata McGraw Hill Edition, 2019 4. Andrew S. Tanenbaum, "Modern Operating Systems", Fourth Edition, Global Edition, Pearson, 2015. 5. William Stallings, "Operating Systems: Internals and Design Principles", Pearson Education, Sixth Edition, 2018.	6. Charles Crowley, "Operating Systems: A Design-Oriented Approach", Tata McGraw Hill Education, 2017 7. https://nptel.ac.in/courses/106/105/106105214/ 8. https://nptel.ac.in/courses/106/106/106106144/ 9. https://nptel.ac.in/courses/106/102/106102132/ 10. https://onlinecourses.nptel.ac.in/noc21_cs44/preview 11. https://nptel.ac.in/courses/106/105/106105172/
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	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)						Summative Final Examination (40% weightage)
		Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)				
		Theory	Practice	Theory	Practice			
		Theory	Practice	Theory	Practice			
Level 1	Remember	20%	-	-	-	20%	-	
Level 2	Understand	40%	-	-	40%	40%	-	
Level 3	Apply	20%	-	-	40%	20%	-	
Level 4	Analyze	20%	-	-	10%	10%	-	
Level 5	Evaluate	-	-	-	10%	10%	-	
Level 6	Create	-	-	-	-	-	-	
	Total	100 %		100 %		100 %		

Course Designers

Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1. Mr.T.Madhan, Team Leader, Tata Consultancy Services, siruseri Campus, Chennai, madhan.tk@gmail.com	1. Dr. S. Janakiraman, Associate Professor, Pondicherry University, sj.dbi@ponduni.edu.in	1. Dr. N. Prasath, SRMIST
2. Mrs.K.Saranya, IT Analyst, Tata Consultancy Services, siruseri Campus, Chennai, saranya.k6@gmail.com	2. Dr. R.Shyamala, Associate Professor, Anna University College of Engineering, Tindivanam, vasuchaaru@gmail.com	2. Dr. M. Eliazar, SRMIST